Electrical design review: Moon reading light, main board + LED board (rev 0.1)

Reviewer: independent electrical review, 2026-11-02. No repo files were edited.

Scope and inputs

- **Sources:** `hardware/main-board/design.py`, `hardware/led-board/design.py`, `hardware/parts.py`, `hardware/lib/make_lib.py` and `hardware/lib/make_footprints.py`.
- **Generated files:** `*.kicad_sch`, `*.kicad_pcb` and `*.kicad_pro`, plus `reports/*.net` and `reports/{erc,drc,route}.json`.
- **Firmware** (pin and address assumptions only): `firmware/include/{config.h,board_map.h}`, `firmware/src/main.cpp`, `firmware/platformio.ini` and `firmware/lib/core/lightcore.h`.
- **Datasheets:** `.../scratchpad/ds/*.txt|pdf`. Diagrams were checked against page renders, not only the pdftotext output.
- **Downloaded for this review:** Vishay S1A–S1M, Doc. 88711, rev. 07-May-2024, from `https://www.vishay.com/doc?88711`, saved at `.../scratchpad/dl_s1j/s1a.pdf`.
- **Not obtained:**
  - Littelfuse SMF5.0A datasheet. littelfuse.com returns 403, and this turn's web-search budget was used up.
  - Samsung DC-bias data for CL21A106KOQNNNE, the 0805 10 µF part actually fitted.
  - The battery's PCM and cell datasheets. Any figure that depends on these is marked **UNVERIFIED** or **ASSUMED**.

Scripted checks (in `.../scratchpad/scripts/`):

- `parity.py`: design.py pin→net, then the netlist, then the PCB pad nets. Also value, footprint and every field (MPN, Farnell and so on) against `parts.py`. **0 mismatches on both boards.** The generated files are current with the sources.
- `fpcmp.py`: every placed footprint compared with its library `.kicad_mod` (pad numbers, positions, sizes and types). **All 56 are identical.**

Severity scale:

- **BLOCKER:** the board won't work, or it is unsafe.
- **MAJOR:** it works badly or is out of spec.
- **MINOR**
- **NOTE**

Findings

ID	Sev	Location	Finding	Evidence	R
B1	**BLOCKER** (firmware/hardware mismatch)	`firmware/include/board_map.h` vs `hardware/led-board/reports/led-board.net` (U1 OUT0–15)	**`OUT_GROUP[]` is stale. 15 of 16 entries are wrong.** See the table below for the current map. The firmware enables OUT4/OUT5, which have no LED, and never enables OUT14/OUT15 (D10 green and blue). Effect: "Warm white" lights D10-red, D8-CW, D7-WW and D6-CW, and every other mode is equally scrambled. `board_map.h` says it is "GENERATED by hardware/build.py led-board", but no generator exists: `grep` finds none in build.py, fab.py or layout.py. board_map.h dates from 16:31; led-board design.py and the PCB date from 16:59.	Netlist nets: `K_D10_R`→U1.6 (OUT0), `K_D8`→U1.7 (OUT1), ..., `K_D10_B`→U1.23 (OUT15). `design.out_group()` evaluates to `[2,1,0,1,-1,-1,0,2,3,4,1,0,1,0,3,4]`. board_map.h has `{0,0,0,0,1,1,1,1,2,3,4,2,3,4,-1,-1}`.	Se {2 Be (w a o

ID	Sev	Location	Finding	Evidence	R
M1	MAJOR	LED board, U1 outputs ↔ D9/D10 green and blue (also worst-bin whites); VLED = VSYS	**Green and blue cannot hold 25 mA over most of the battery range. Worst-VF parts never regulate, even on USB.** The TLC59116 needs VLED ≥ VF + knee. From cell to LED board, the path drops about 0.15 V at 0.25–0.38 A: cell/PCM, Q3+Q1+Q2 (about 3 × 45 mΩ) and the 200 mm 28 AWG pair plus contacts (about 0.13 Ω). See the regulation table below. Below these voltages the current falls and the colour of the mixed modes drifts. Affected modes: 3000 K and 4000 K (green trim), 6500 K (G+B), Candlelight (R + 90 % G), and the Green and Blue modes. Brightness also sags. At low CG (dim levels) the knee and VF fall, so the problem is mainly at high brightness.	`tlc59116.txt` l.404: "IO(1) … VO = 0.6 V, Rext = 720 Ω … 26 mA" is the only guaranteed headroom point. Fig. 2 (p9 render) shows a knee of about 0.35–0.4 V at 20 mA and 0.45–0.6 V at 40 mA. `we150141m173100.txt` l.121–130: VF G/B 3.2 typ / **3.6 max** @20 mA; R 2.0/2.4. The p3 curve at 25 mA reads R ≈ 2.10, B ≈ 3.20, G ≈ 3.32 V. Bridgelux DS312 Table 1 (l.132–136): VF 2.90/3.13/3.30 V @150 mA; Fig. 9 reads about 2.74 V @25 mA.	D di (a fr (b gi 3. (c (d cc ar 3. D ei
M2	MAJOR	Main board Q2/SW1 (only power control); LED board U1 VCC, RESET tied to VLED	**The MCU cannot remove VSYS/VLED or put the LED driver in a guaranteed low-current state.** After firmware decides the battery is empty (`sleep_until_resumed()`, VSYS < 3.3 V) with the slide switch left ON, the TLC59116 stays powered. Its only datasheet supply-current limit is **25 mA max (outputs off, Rext open) / 29 mA (Rext 720 Ω), at VCC 5.5 V**. No standby or typical figure is given. MCU power-down (≤ 2 µA), LM75 shutdown (≤ 1 µA) and the Q2 gate divider (VSYS_RAW / 1.01 MΩ ≈ 4 µA) add to it. The cell is therefore run down to the pack PCM's over-discharge cut-off (threshold ASSUMED 2.4–3.0 V, PCM unknown), which relies wholly on the PCM. The same TLC ICC is a fixed overhead while the lamp is on.	`tlc59116.txt` l.438–450: ICC "OUT0 to OUT15 = OFF, Rext = Open … 25", "Rext = 720 Ω … 29" mA max at VCC 5.5 V. l.974: "Standby mode occurs when all output channels are disabled … or by pulling the RESET pin low" (no current figure). `main.cpp sleep_until_resumed()` only turns off the oscillator (MODE1 OSC = 1).	Ei (a of to sv (b pl ke M =
M3	MAJOR (safety, foreseeable misuse)	Main board J4 pin 1 = VSYS (silk "UPDI: 1=VSYS …")	**A programmer or USB-UART that supplies VCC on J4.1 back-feeds the LiPo with no current or voltage control.** Switch OFF with a 5 V programmer: VSYS 5 V → Q2 body diode (D→S) → VSYS_RAW ≈ 4.4 V. Q1 is ON (gate = VBUS = 0 via R5, Vgs ≈ −4.4 V) → BAT → Q3 ON → cell, charged from about 4.4 V. Switch ON: Q2's channel conducts, so it is the same path with even less resistance. The cell is pushed above 4.20 V until the PCM over-charge trips (~4.25–4.3 V, ASSUMED). With a 3.3 V programmer and the switch ON, the cell (3.7–4.2 V) instead back-drives the programmer's 3.3 V rail through Q2/Q1/Q3.	DMG2305UX p1 internal schematic: body diode anode at D, cathode at S (render checked). Netlist: `J4.1`, `Q2.3(D)` and `U2.1` on VSYS. `Q1.1(G)` = VBUS with R5 100 k to GND.	M su m J4 ta U pr di V

ID	Sev	Location	Finding	Evidence	R
M4	MAJOR (USB spec; low practical risk with ≥ 1 A chargers)	Main board J1/R1/R2 (Rd-only sink), U1 (417 mA), D3 (lamp load from VBUS)	**VBUS draw can exceed the 500 mA allowed for "Default USB Power".** Charging takes 417 mA nominal (375–458 mA with the ±10 % IREG tolerance at 2 k). The lamp load through D3 adds about 0.2–0.4 A (TLC ICC + LEDs + MCU). Total VBUS current is about 0.6–0.85 A. The sink presents Rd only and never reads the CC voltage, so it cannot know the source capability. On a USB 2.0 host port, or a source advertising Default power, this is out of spec and may trip the port's limit. VBUS then sags, and when VSYS_RAW < BAT − 0.6 V the Q1 body diode starts carrying load from BAT, so charge termination is never reached.	`an1149.txt` l.362–375: "…assured not to exceed the 500 mA limit…", "The USB Specification clearly defines … up to 500 mA loads". `mcp73831.txt` l.184: IREG 450/505/550 mA at 2.0 k, i.e. about ±10 %.	Ei (a pi w (b ał ≤ (c ch
m1	MINOR	Main board Q1 gate = VBUS, R5 100 k, C1 10 µF	**Slow handover on USB unplug.** After unplug, C1 drains into the load through D3 until VBUS ≈ VSYS_RAW + 0.4 V. From then on it decays only through R5: τ = 100 k × ~6–8 µF effective ≈ 0.6–0.8 s. Q1 needs VBUS ≈ VSYS_RAW − 1 V to turn on, which takes about 0.3–0.4 s. In that window the load runs through the Q1 body diode (Vsd ≈ 0.6 V at 0.25–0.4 A), so VSYS ≈ BAT − 0.6 V. With BAT ≤ 3.6 V, VSYS falls below the TLC59116 VCC minimum of 3.0 V, and firmware may see a false "empty" (`VBAT_EMPTY_MV` 3300). Expect a visible brightness and colour dip.	DMG2305UX Fig. 8 (p4 render): about 0.6 V at 0.3 A. `tlc59116.txt` l.355: VCC min 3 V.	Cl fr gi er ti
m2	MINOR	Main board Q3/R10 with USB present	**Reverse-battery protection is defeated while USB is connected.** With a reversed pack, BAT_RAW ≈ −3.7 V. The charger drives BAT positive, so Q3's Vgs = −V(BAT) turns it on whenever V(BAT) > |Vth| (0.5–0.9 V). The circuit settles in pre-conditioning at about 42 mA (10 % IREG): about 0.2 W in Q3 and about 0.2 W in U1. The reversed cell is discharged at about 42 mA until its PCM cuts, and the red "charging" LED lights. Nothing is destroyed. Without USB, Q3 blocks correctly.	MCP73831 option AC: IPREG/IREG 10 % (p26 table). Flowchart (l.891): "VBAT < VPTH → PRECONDITIONING … STAT = LOW".	Ac re bi to
m3	MINOR	Main board D1 SMF5.0A vs U1 VDD	**The TVS clamp is above the MCP73831 absolute maximum.** VDD abs max is 7.0 V and operating max is 6 V. A 5.0 V-standoff TVS has VBR ≥ 6.4 V and, typically for this voltage class, VC ≈ 9.2 V at IPP (**UNVERIFIED**, datasheet not obtained). Hot-plug ringing of 10 µF ceramic plus cable inductance can overshoot 5 V by up to 2×. The TVS will not hold VDD ≤ 7 V during a surge.	`mcp73831.txt` l.142: "VDD … 7.0V" abs max; l.157: VDD 3.75–6 V.	G lo 7- (o ri
m4	MINOR	Main board U1 thermal; no cell-temperature qualification	Worst case, U1 dissipates (5.5 − 2.79) × 0.458 A ≈ 1.24 W, and typically 0.5–0.8 W. In SOT-23-5 (θJA 130–230 °C/W) this will run in thermal regulation at about 110–150 °C die. D3 adds about 0.35 W (0.4 A × 0.88 V) while the lamp runs from USB. All of this sits inside the pack, beside the LiPo. The MCP73831 has no NTC input, so nothing stops charging of a hot or cold cell.	`mcp73831.txt` l.1234–1252: "PowerDissipation = (VDDMAX − VPTHMIN) × IREGMAX … 1.54W … will cause thermal regulation". Thermal table: SOT-23 θJA 230 °C/W (min Cu), 130 (large Cu). S1J Fig. 3 (p3 render): about 0.88 V at 0.4 A.	Ke cc IR w pa

ID	Sev	Location	Finding	Evidence	R
m5	MINOR	LED board D9/D10 (Würth) at 25 mA	IOUT = 25.0 mA ± 8 %, so up to 27 mA. The Würth derating curve allows 30 mA only up to 50 °C ambient, falling linearly to 10 mA at 85 °C. 27 mA is therefore permitted only to about 55 °C ambient. The LEDs share a 26 mm board inside the moon with 8 whites and a TLC dissipating 0.2–0.6 W.	`we150141m173100.txt` l.12: IF 30 mA. Derating curve (p4 render): 30 mA to 50 °C, then 10 mA at 85 °C. `tlc59116.txt` l.405: ± 8 %.	H th ab te
m6	MINOR	LED board U1 pin 25 RESET hard-tied to VLED	There is no hardware reset or standby control. POR only re-arms if VCC falls below 0.2 V. A sag (USB unplug m1, cable glitch, brief switch toggle) that leaves VCC between 0.2 and 2.5 V can leave the driver in an undefined state. Firmware only re-initialises when an I²C write fails.	`tlc59116.txt` l.795–798: "…Thereafter, VCC must be lowered below 0.2 V to reset the device." VPOR 2.5 V (l.400).	R k Al (0 pe
m7	MINOR	Main board silk "+" at (1.2, 52.5) for J2	The only battery-polarity marking lies **inside the J2 housing outline** (F.Fab x 0.45–8.15, y 50.5–58.5), so it is hidden once the connector is fitted. The "+ = pin 1" claim itself is consistent.	`pcbnew` dump: J2 Fab bbox x 0.45..8.15, y 50.50..58.50; text '+' at (1.2, 52.5).	M at "-
m8	MINOR	Main board (no VBUS / STAT / BAT sense to MCU)	Firmware's "battery voltage" is VDD = VSYS. On USB it reads about 4.0–4.3 V (VBUS − D3), so firmware can see neither the charge state nor USB presence. That blocks fixes for M4 and m1. 11 GPIOs are free.	Netlist: `CHG_STAT` goes only to D2/U1; VBUS has no MCU pin. `main.cpp` `vdd_mv()` measures VDD.	A a to th sv
N1	NOTE	Main board Q2 soft-start (R7 10 k, C5 100 nF G–S)	The gate RC is about 1 ms, but C5 sits gate–source (no Miller dv/dt control). Simulated inrush into about 14–21 µF on VSYS (C4 + LED-board C2 + MCU caps) is **0.7–1.3 A peak for about 0.1 ms**, about 0.3–0.4 ms after closing. It is within the DMG2305UX IDM of 15 A and the JST-SH 1 A short-term, and below typical PCM over-current delays. Acceptable.	Square-law model, Vth 0.5–0.9 V (DMG2305UX VGS(th) −0.5…−0.9 V, l.114), K ≈ 2–5 A/V².	O fo w
N2	NOTE	Main board Q1 gate directly on VBUS	Vgs normally stays within +0.7… +5 V (VBUS − VSYS_RAW) or −4.2 V, so it is within ±8 V. An ESD hit on VBUS, clamped by the TVS at its VC while the battery holds VSYS_RAW at about 4 V, could exceed +8 V for nanoseconds.	DMG2305UX l.82: VGSS ±8 V.	A (w tr
N3	NOTE	Main board C1/C2 = CL21A106KOQNNNE (10 µF 16 V X5R 0805)	The MCP73831 needs ≥ 4.7 µF on VDD and VBAT. The DC-bias curve for this 0805 part is **not in the folder (UNVERIFIED)**. The 0603 10 V part, rightly rejected, keeps only 2.85 µF at 5 V. Stacking −10 % tolerance, about −45 % at 5 V (typical for this class, ASSUMED) and −15 % X5R at 85 °C gives about 4.2 µF. That is marginal for the "no battery" stability case.	`mcp73831.txt` l.844–852: "Bypass to VSS with a minimum of 4.7 µF … loop stability when the battery is disconnected". KEYSPECS §10 (0603 data).	Pu Cl 2
N4	NOTE	Main board D1 footprint `D_SOD-123F`	The pads (1.1 × 1.1 at ±1.4) are smaller than onsemi's SOD-123FL recommendation (1.25 × 1.22 at ±1.475). The Littelfuse SMF land pattern could not be checked (datasheet not obtained). Polarity is correct: pad 1 = K = VBUS.	KEYSPECS §6–7; footprint file.	Cl dr
N5	NOTE	Main board SW1 PCM12	There is no ON/OFF silk. Position 1 (1–2: common to GND) = ON; position 2 (2–3: common to VSYS_RAW) = OFF. Which knob direction is position 1 is not stated in the drawing text.	`pcm12.txt`: "POS. 1 … 1-2, POS. 2 … 2-3".	Ve th

ID	Sev	Location	Finding	Evidence	R
N6	NOTE	Main board J4.3 TX	A USB-UART adapter left connected while the switch is OFF can back-power the MCU and LED board through its RX pull-up into PA1's ESD diode.	Datasheet abs max: pin ≤ VDD + 0.5 V (KEYSPECS ATtiny).	D se
N7	NOTE	Main board U1/D2 with no battery and USB present	The MCP73831 cycles between "no battery" and "charge", so the red LED flickers. Expected behaviour.	`mcp73831.txt` §4.2 Battery Detection (l.895–905).	M
N8	NOTE	Interconnect: cable LC	The 200 mm VLED/GND pair (about 0.1–0.2 µH loop) with the LED-board 10 µF (about 6 µF effective) resonates near 130–180 kHz, close to the 97 kHz PWM fundamental and its harmonics. Q ≈ 2 (cable R about 0.09 Ω). Expect some VLED ripple, which also lands on the MCU VDD that is used as the ADC reference.	Estimate (ASSUMED wire L).	Sc ac or
N9	NOTE	LED board `.kicad_pcb`	Board thickness in the file is **1.6 mm**, but `led-board/design.py` says "1.0 mm FR4". Nothing in build.py or fab.py sets it, so the STEP and fab notes will be wrong for the optics stack.	`led-board.kicad_pcb` l.6 `(thickness 1.6)`.	Se do
N10	NOTE	ERC/DRC reports	Main ERC: 0 violations. LED ERC: 1 `lib_symbol_mismatch` (LM75B). That one is benign: the library `LM75B` *extends* `LM75C` and kigen embeds the flattened copy; the pins are identical (1 SDA, 2 SCL, 3 O.S., 4 GND, 5 A2, 6 A1, 7 A0, 8 +Vs). DRC: 0 errors, 0 unconnected, parity 0. Warnings are silk only: SW1 silk clipped by the board edge, which is intended; LED-board silk overlaps; "MOON LED v0.1" clipped by mask; a 0.6 mm text height on "1 GND 2 V 3 SDA 4 SCL", below the 0.8 mm board minimum and probably below fab legibility. The ignored DRC rules (`footprint_filters_mismatch`, `footprint_type_mismatch`, `missing_courtyard`, `npth/pth_inside_courtyard`) are KiCad 9 defaults. There are no DRC or ERC exclusions.	`reports/*.json`; `.kicad_pro`.	Er 0.
N11	NOTE	Main board J3 (Qwiic pinout)	J3 carries VSYS (up to about 4.7 V on USB), not 3.3 V. A stock 3.3 V Qwiic device would be over-voltaged. This is already noted in the schematic comment 2.	—	Ke
N12	NOTE	Battery J2 "+ = pin 1"	Internally consistent: J2.1 = BAT_RAW; the footprint's pin 1 = JST circuit 1 (the JST SM4 side-entry drawing shows circuit 1 rightmost with the mating side up, and KiCad's pad 1 is the same rotated 180°); the "+" silk is on the pin-1 side. There is **no industry convention** for JST-PH LiPo wiring, so the pack must match. Q3 protects without USB (see m2 with USB).	`jst_ph.pdf` p1 (render): "Circuit No.1".	Sp or

B1 detail: TLC59116 output map

OUT	Pin	Netlist load	Correct group	board_map.h
0	6	D10 R	2 R	0 WW ✗
1	7	D8 (5000 K)	1 CW	0 ✗
2	8	D7 (2700 K)	0 WW	0 ✓
3	9	D6 (5000 K)	1 CW	0 ✗
4	11	— (NC)	−1	1 ✗
5	12	— (NC)	−1	1 ✗
6	13	D5 (2700 K)	0 WW	1 ✗
7	14	D9 R	2 R	1 ✗
8	15	D9 G	3 G	2 ✗

OUT	Pin	Netlist load	Correct group	board_map.h
9	16	D9 B	4 B	3 ✗
10	17	D4 (5000 K)	1 CW	4 ✗
11	18	D3 (2700 K)	0 WW	2 ✗
12	20	D2 (5000 K)	1 CW	3 ✗
13	21	D1 (2700 K)	0 WW	4 ✗
14	22	D10 G	3 G	−1 ✗
15	23	D10 B	4 B	−1 ✗

M1 detail: regulation limits at 25 mA, 25 °C

Assumptions:

- TLC knee: 0.40 V typical (Fig. 2) and 0.60 V guaranteed (IO(1) test point).
- VLED ≈ OCV − 0.15 V. This covers the cell, PCM, 3 FETs and cable at 0.25–0.38 A (cell and PCM resistance ASSUMED 0.15–0.25 Ω).
- The SoC figures use a generic LiPo OCV curve (ASSUMED).

Colour	VF @ 25 mA: typ / worst	VLED needed: typ / worst	Cell OCV needed: typ / worst	Consequence
Red (Würth)	2.10 / ~2.45 V	2.50 / 3.05 V	always OK above the firmware's 3.3 V cut	regulated
White (Bridgelux, both CCTs)	2.74 / ~2.91 V (VF bin max 3.30 vs typ 3.13 @150 mA)	3.14 / 3.51 V	3.30 / 3.66 V	Typical parts regulate almost to empty. Worst-bin parts dim over the last ~10–15 %.
Blue (Würth)	3.20 / ~3.65 V	3.60 / 4.25 V	3.75 / **4.40 V (unreachable)**	Typical: regulated only above ~35 % SoC. Worst case: never on battery; marginal on USB (VSYS ≈ 4.05–4.35 V).
Green (Würth)	3.32 / ~3.70 V	3.72 / 4.30 V	3.87 / **4.45 V (unreachable)**	Typical: regulated only above ~55–60 % SoC. Worst case: never on battery, and not on USB under load.

Power-path verification (all states)

The P-FETs as built are `Q_PMOS_GSD` on the KiCad SOT-23 footprint: pad 1 = G, pad 2 = S, pad 3 = D. That matches the Diodes DMG2305UX top view: G bottom-left, S bottom-right, D top. The body diode runs D→S (anode at drain), confirmed on the p1 render.

FET	G	S	D	Body diode	Role
Q3	Q3_G (R10 10 k → GND)	BAT	BAT_RAW (J2.1)	BAT_RAW → BAT	reverse-battery (correct orientation)
Q1	VBUS (R5 100 k → GND)	VSYS_RAW	BAT	BAT → VSYS_RAW	AN1149 load share (correct orientation)
Q2	SW_GATE (R6 1 M to S, C5 100 n G–S, R7 10 k to SW1 common)	VSYS_RAW	VSYS	VSYS → VSYS_RAW	load switch (correct orientation)

D3 S1J: pad 1 = K = VSYS_RAW, pad 2 = A = VBUS. Correct.

State	Q3	Q1	Q2	Result / Vgs check
Battery only, switch ON	ON (Vgs = −VBAT, −3.0… −4.2 V)	ON (Vgs = −VSYS_RAW)	ON (gate ≈ VSYS_RAW × 10k/1.01M ≈ 40 mV)	VSYS ≈ VBAT − I × (3 × ~45 mΩ). All |Vgs| ≤ 4.2 V, within ±8 V.
Battery only, switch OFF	ON	ON	OFF (G = S = VSYS_RAW via R7 and R6, no current)	VSYS isolated. **VSYS_RAW and BAT stay live.** Standby budget below.
USB only, switch ON	ON (charger output drives BAT_RAW, no cell)	OFF (Vgs = VBUS − VSYS_RAW ≈ +Vf(D3))	ON	VSYS = VBUS − Vf(D3) − ε: ≈ 4.3 V light load, ≈ 4.1 V at 0.4 A (S1J ≈ 0.88 V at 0.4 A).
USB + battery, switch ON	ON (charge current S→D)	OFF	ON	Load fed from VBUS through D3. The charger sees only the cell, so termination works (AN1149). The body diode conducts only if VSYS_RAW < BAT − ~0.5 V (weak USB, see M4).
USB, switch OFF	ON	OFF	OFF	Charges normally. VSYS ≈ 0. No MCU pin is driven: pull-ups go to VSYS, the LED board is on VSYS, buttons go to GND. **No back-powering.** Q1 Vgs ≈ +0.7 V.

State	Q3	Q1	Q2	Result / Vgs check
Reversed battery, no USB	OFF (Vgs ≥ 0; body diode reverse-biased, IDSS ≤ 1 µA)	—	—	Blocked. BAT_RAW is reached only by Q3.D and TP2.
Reversed battery + USB	partly ON	OFF	—	About 42 mA flows (see m2).
USB unplug under load	ON	ON after ~0.3–0.4 s (see m1)	ON	Body-diode handover dip.
Programmer powering J4.1	ON	ON	body diode or ON	Back-charges the cell (see M3).

Inrush (Q2): about 1 A peak for ~0.1 ms. Acceptable (N1).

Battery hot-plug with the switch ON: C5 drags the gate up together with the source, so Q2 starts OFF and then soft-starts. Good.

Switch-OFF battery drain (battery only)

Leakage path	Datasheet basis	Worst case	Typical
U1 VBAT reverse leakage (VDD < VSTOP; R5 holds VDD at 0)	`mcp73831.txt` l.228: 0.15 typ / 2 µA max	2 µA	0.15 µA
Q2 IDSS (VSYS_RAW→VSYS, VDS ≈ −4 V)	DMG2305UX l.111: ≤ 1 µA @ −20 V	≤ 1 µA	~nA
D3 reverse (VSYS_RAW→VBUS→R5)	S1J IR ≤ 1 µA @ 600 V; Fig. 4 shows ~0.004 µA @ 10 % VRRM	≤ 1 µA	< 0.01 µA
C2 and C3 IR (10 µF each at 4.2 V)	Samsung spec ≥ 100 MΩ·µF, i.e. 10 MΩ (ASSUMED the same for the 0805 part)	0.84 µA	nA
Q1 and Q3 gate leakage (through R5/R10)	IGSS ±100 nA @ 8 V	0.2 µA	nA
R6/R7/C5	0 V across them in OFF	0	0

Total: ≤ ~5 µA worst case, about 0.2–0.3 µA typical. That is negligible: the pack's own PCM (typically a few µA) dominates. The TVS leakage is not involved, because VBUS stays at about 0 V.

Checked and found correct (coverage)

Netlist, footprint and source parity

- design.py ↔ netlist ↔ PCB pad nets: 0 mismatches on both boards.
- All MPN, value and footprint fields equal `parts.py`.
- All 56 placed footprints are identical to their library sources (pads, positions, sizes).

Main board

- **J1 GCT USB4125-GF-A** (`USB_C_Receptacle_GCT_USB4125-xx-x_6P_TopMnt_Horizontal`):
  - Pads left to right B12, B9, A5, B5, A9, A12 match the GCT "Recommended PCB Layout" (usb4125.pdf render).
  - A5 = CC1 and B5 = CC2, each with its own 5.1 k Rd (R1, R2). Separate Rd per CC is correct.
  - A9/B9 = VBUS; A12/B12 and shell S1 = GND.
  - The 1.00 mm stake variant matches the blank suffix.
- **D1 SMF5.0A:** pad 1 = K → VBUS, A → GND. Unidirectional orientation is correct. Electrical values are UNVERIFIED (m3, N4).
- **U1 MCP73831-2ACI/OT:**
  - SOT-23-5 pins 1 STAT, 2 VSS, 3 VBAT, 4 VDD, 5 PROG match datasheet Table 3-1, the KiCad symbol and the footprint numbering.
  - PROG 2.4 k gives IREG = 1000 V / 2.4 kΩ = 416.7 mA (0.35 C for 1200 mAh).
  - Option AC (p26): IPREG 10 % = 42 mA; VPTH 66.5 % = 2.79 V; ITERM 7.5 % = 31 mA (23–39 mA); VRTH 96.5 % = 4.05 V; VREG 4.20 V ± 0.75 %.
  - C1 and C2 are both 10 µF nominal, ≥ 4.7 µF (DC-bias caveat N3).
  - The battery-detection impedance seen at VBAT is > 7 MΩ: Q1 is off with VBUS present, Q3's open drain goes only to TP2.
- **D2 + R4 (STAT LED):**
  - Kingbright APT1608SURCK pin 1 = cathode (p1 symbol "1 —◁— 2"); KiCad LED_0603 pad 1 = K → CHG_STAT; anode → R4 1 k → VBUS.
  - The LED is ON while STAT = L (charging) and OFF at H (complete) or Hi-Z.
  - LED current ≈ (5 − 1.8 − 0.4)/1 k ≈ 2.8–3 mA, well within ISINK 25 mA.
- **Q1, Q2, Q3 DMG2305UX:**
  - GSD pinout and body-diode orientation are correct; all Vgs stay within ±8 V in normal states.
  - RDS(on) ≤ 100 mΩ at Vgs = −2.5 V, which covers the battery down to 2.5 V.
- **D3 S1J:**
  - SMA pad 1 = K (Vishay: "color band denotes cathode end").
  - VF ≈ 0.88 V at 0.4 A and ≈ 0.65 V at 10 mA (Fig. 3). Reverse leakage is in the nA range at 4 V. The parts.py claim is verified.

    - About 0.35 W in SMA (RθJA 75 °C/W with 5 × 5 mm pads) is OK.
- **SW1 PCM12SMTR:**
    - KiCad pads 1 (−2.25), 2 (+0.75) and 3 (+2.25) match the C&K pitch of 3.0/1.5 mm.
    - Pad 2 = common = SW_COM. Position 1 (1–2) = GND = ON; position 2 (2–3) = VSYS_RAW = OFF.
    - The switch carries only gate current (µA) against a 0.3 A rating.
- **SW2–SW4 PTS645:** C&K pins 1–2 and 3–4 are internally connected (SMT schematic, p3 render). KiCad numbers both top pads "1" and both bottom pads "2", which is consistent with the 2-pin SW_Push. Buttons go to GND, with internal pull-ups on PA6/PA5/PA4.
- **J2 JST S2B-PH-SM4-TB:** pin numbering is consistent with JST circuit 1 (N12); rated 2 A.
- **J3 JST SM04B-SRSS-TB:** Qwiic order 1 GND, 2 VSYS, 3 SDA, 4 SCL, matching the LED-board pads 1–4. Rated 1 A against a ~0.4 A load.
- **U2 ATtiny3216-SN (SOIC-20W):**
    - Pin 1 VDD and pin 20 GND; 16 PA0/UPDI → J4.2; 17 PA1 → TX (J4.3, USART0 alternate TxD per Table 5-1 footnote 3, `Serial.swap(1)`).
    - 10 PB1 = SDA and 11 PB0 = SCL (TWI0 default, Table 5-1).
    - 2/3/4 = PA4/PA5/PA6 = DOWN/MAIN/UP, which agrees with config.h. Unused pins are NC and set to INPUT_PULLUP in firmware.
    - 10 MHz (20 MHz osc / 2) with BOD 2.6 V is explicitly ensured. Table 36-3 note 2: "Operation ensured down to BOD triggering level, VBOD with BODLEVEL2". VSYS max ≈ 4.9 V < 5.5 V.
    - Decoupling 100 n + 1 µF at the VDD pin.
    - No MCU pin can be driven above VDD in any switch/USB state (except N6 with an external adapter).
- **I²C:**
    - 3.3 k pull-ups to VSYS give 1.3 mA at 4.3 V, under the 3 mA standard-mode sink limit.
    - Bus capacitance ≈ 50–80 pF: MCU ~10 pF, TLC Cio 8 pF / Ci 5 pF, LM75, ~8 pF of traces, a 200 mm cable ~10–20 pF, test pads. tr ≈ 0.85 × 3.3 k × 80 pF ≈ 0.22 µs, under 1 µs at 100 kHz and also under 0.3 µs for 400 kHz.
    - All devices share one rail (VSYS = VLED − cable drop of ~15–30 mV). TLC VIH = 0.7·VCC and LM75 VIH = 0.7·VCC are met. TLC VI ≤ VCC + 0.4 V is met.

LED board

- **U1 TLC59116IPWR:**
    - The custom symbol pins match TI's PW pinout exactly: 1 REXT, 2–5 A0–A3, 6–9 OUT0–3, 10 GND, 11–14 OUT4–7, 15–18 OUT8–11, 19 GND, 20–23 OUT12–15, 24 GND, 25 /RESET, 26 SCL, 27 SDA, 28 VCC. `PIN_OF_OUT` in design.py matches too.
    - TSSOP-28 footprint OK. The PW package has no thermal pad.
    - A0–A3 go to GND, giving address 0x60 = `110 0000` (I.1532). That is distinct from All-Call 0x68 (D0h, enabled at POR), SWRST 0x6B (D6h) and SUBADR 0x69/0x6A/0x6C (disabled at POR), and from the LM75 at 0x48. Firmware writes MODE1 = 0, which also disables All-Call.
    - RESET tied to VCC is allowed (I.803 "requires a pullup … if no active connection"). See m6 for the trade-off.
    - Rext 750 Ω gives IOUT = 1.26 × 0.992 × 15 / 750 = **25.0 mA** with the default IREF 0xFF (CG 0.992).
    - VCC = VSYS stays 3.3–4.9 V, inside 3.0–5.5 V. Firmware's empty threshold keeps VCC ≥ 3.3 V.
    - Unused OUT4/OUT5 may float (I.1966: "unused outputs can be left floating without issue").
    - Decoupling: C1 100 n at pin 28 plus 10 µF bulk.
- **D1–D8 Bridgelux 2835:**
    - The datasheet marks the large pad "−" (Fig. 16 bottom view). KiCad LED_PLCC_2835 pad 1 (2.2 × 2.2, large) = K → TLC output; pad 2 = A → VLED.
    - 25 mA is far below IF max 150 mA. No reverse bias occurs in this topology: anodes are always ≥ cathodes, and the open-drain outputs cannot source current.
    - The KiCad pads are slightly larger than Bridgelux's (4.0 vs 3.74 mm overall), which is acceptable.
- **D9/D10 Würth 150141M173100:**
    - Pins 1 = common anode, 2 = B-K, 3 = R-K, 4 = G-K (p1 schematic render). This equals KiCad `Device:LED_ABRG` (1 A, 2 BK, 3 RK, 4 GK) and the footprint pads.
    - Pad layout (3 TL, 1 TR, 4 BL, 2 BR) matches the land pattern.
    - The KiCad silk and fab chamfer sit at pad 3, the same corner as Würth's "Polarity Mark (Cathode; red LED)". No orientation trap.
    - 25 mA against 30 mA abs max (see m5). Red Pdiss 2.1 V × 27 mA = 57 mW < 72 mW; green 92 mW < 108 mW.
- **U2 LM75BDP (TSSOP-8, SOT505-1):**
    - NXP Fig. 3 TSSOP8 pinout: 1 SDA, 2 SCL, 3 OS, 4 GND, 5 A2, 6 A1, 7 A0, 8 VCC. This matches the KiCad LM75B symbol (via LM75C) and the `TSSOP-8_3x3mm_P0.65mm` numbering.
    - Footprint pads at ±2.15 (1.1 × 0.4) give 5.4 mm outer / 3.2 mm inner, against the NXP reflow land of 5.5 / 3.2 mm (Fig. 28). OK.
    - A0–A2 go to GND, giving 0x48 (`1001 000`). The address pins are not floating, as required (I.353).

    - OS is an open-drain output (I.210), so leaving it NC is fine. Its default comparator mode at 80 °C has no effect.
    - VCC range 2.8–5.5 V is OK. Decoupling C3 100 n.
- **J1 wire pads:** 1 GND, 2 VLED, 3 SDA, 4 SCL, the same as main-board J3. The pin-1 silk bar is present.

Firmware ↔ hardware

- config.h matches the hardware: TWI default pins, ADDR_TLC 0x60, ADDR_LM75 0x48, buttons PA6/PA5/PA4, `SERIAL_SWAP 1` (TX = PA1 = J4.3), and the platformio settings 10 MHz / BOD 2.6 V.
- The only hardware/firmware disagreement is `board_map.h` (B1).

Reports

- ERC/DRC were re-read and project settings checked for hidden exclusions and ignores (N10).
- The routes report shows no failed nets on either board.